Sovereign Risk, Financial Intermediation, and Stock Returns

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Abstract

Using Eurozone data from 2005 to 2013, we find that the stock prices of firms vulnerable to financial intermediation disruption are particularly sensitive to changes in sovereign credit spreads. Such higher sensitivity is fully due to forward P/E ratios and not due to short-term expected earnings, indicating that stocks are priced as if sovereign risk has effects on vulnerable stocks beyond the short-term. Estimation of a structural model indicates that such longer-term effects are expected to be large. Our findings provide novel support for recently developed theories of sovereign default proposing that financial intermediation disruption is a key cost of sovereign default.

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1 Introduction

Sovereign debt is fundamentally different from corporate debt because it is not enforceable in a court of law. Why are lenders willing to extend uncollateralized loans if they cannot seize assets upon default? It must be because borrowers face costs of some form as they fail to repay. Traditional theories of sovereign debt propose that the cost arises from creditors' retaliation, such as exclusion from future borrowing and other sanctions. These traditional theories, however, have little empirical support.[1]

Recently, new theories propose that the main cost of sovereign default is disruption in domestic financial intermediation (e.g., Gennaioli et al. 2014a).[2] Disruption occurs because of the following frictions. First, domestic banks hold substantial positions in their sovereign's debt. Second, sovereigns cannot selectively default on foreign bondholders only. Third, domestic firms and consumers rely primarily on domestic banks for finance and cannot easily switch to others sources of capital. Therefore, when sovereign risk increases, domestic banks, distressed by sovereign debt losses, reduce private credit supply, which negatively affects investment and output.[3]

This paper studies a key cross-sectional implication of the new theories of sovereign debt. Such theories predict that firms that are particularly vulnerable to financial intermediation disruption should be more strongly affected by the prospect of sovereign default. Our results confirm this prediction.

[1] For the paucity of empirical support for traditional theories, see Borensztein and Panizza (2009), Panizza et al. (2009), and Levy Yeyati and Panizza (2011).

[2] In addition to Gennaioli et al. (2014), new theories of sovereign debt focusing on collateral damage through the domestic banking system include Basu (2010), Brutti (2011), Sandleris (2014), Mengus (2013), Sosa-Padilla (2014), Bocola (2014), and Perez (2015). Related theories that also feature the financial intermediation channel of sovereign default include Bolton and Jeanne (2011) and Farhi and Tirole (2014).

[3] Note that disruption in domestic financial intermediation may start even before sovereign default, as the value of sovereign debt on banks' balance sheets deteriorates.

We study European stock returns from 2005 to 2013, a period that includes the Euro sovereign crisis. In line with previous research, we find that stock returns are contemporaneously negatively correlated with changes in sovereign spreads.[4] More importantly, we show that non-financial firms more vulnerable to credit market disruption display higher sensitivity to changes in sovereign spreads.

Following the literature, we use corporate asset tangibility to proxy for vulnerability to financial intermediation disruption.[5] Firms with less pledgeable collateral are likely to be more negatively affected by a reduction in aggregate credit supply. We find that firms in the lowest tercile of asset tangibility have sovereign risk betas that are twice as large as other non-financial firms. As new theories predict, firms vulnerable to aggregate credit disruption respond more strongly to sovereign risk changes.

Our results are robust to using two other proxies for vulnerability to financial intermediation disruption. These alternative proxies are firm age (Petersen and Rajan 1994) and the Rajan-Zingales external finance dependence metric (Rajan and Zingales 1998). We find that firms incorporated after 2001 have sovereign risk betas that are double that of older firms. Similarly, the sovereign betas of firms in the highest tercile of the Rajan-Zingales external finance dependence metric are 75% larger than that of other firms.

[4] Many papers find a negative contemporaneous correlation between sovereign credit spreads and stock returns. For example, Bailey and Chung (1995), Andrade (2009), Longstaff et al. (2010), and Dieckmann and Plank (2012).

[5] Holmstrom and Tirole (1998) and Almeida and Campello (2007) provide conceptual justification for the importance of asset tangibility. Papers using asset tangibility to proxy for credit vulnerability include Claessens and Laeven (2003), Braun (2003), Qian and Strahan (2007), Chor and Manova (2012), Manova (2013), Campello and Giambona (2014), and Manova et al. (2014). In Appendix A we show that Eurozone firms with low asset tangibility in fact experience stronger reduction of bank loans when sovereign spreads increase. This finding validates using asset tangibility to proxy for vulnerability to financial intermediation disruption in our particular setting.

To mitigate concerns that omitted variables unrelated to sovereign default drive differences in sovereign risk sensitivity, we also examine short-window returns immediately following policy announcements directly affecting sovereign debt markets. Specifically, we focus on three announcements: the creation of the European Financial Stability Fund in May 2010, the European Central Bank's President Mario Draghi "Whatever-It-Takes" speech in July 2012, and the creation of Outright Monetary Transactions in September 2012. We find that vulnerable firms have higher sovereign risk sensitivity not only in general but also following the policy announcements.

Importantly, further results show that stock prices behave as if sovereign risk has a long-term effect on firms vulnerable to financial intermediation disruption. This matters because the new theories of sovereign default rely on intermediation frictions whose medium- to long-run relevance is unknown.[6] If those frictions are irrelevant beyond the short-term, then it is unlikely that financial intermediation disruption is a key cost of sovereign default. Thus, finding (prospective) long-term effects of sovereign risk provides important support for the new theories of sovereign default.

Specifically, we decompose sovereign risk sensitivity onto immediate and long-term components as follows. First we use I/B/E/S earnings forecasts to decompose stock returns (without dividends) onto changes in 3-year earnings expectations and changes in 3-year forward P/E ratios. Then we separately regress each return component onto contemporaneous changes in sovereign spreads. This effectively decomposes a stock's sovereign risk beta into a short-term earnings beta and a residual

[6] Specifically, if financial intermediation frictions only have short-run effects, then the implementation of positive NPV projects in the defaulting economy would be merely briefly delayed. In the medium-term, firms would find sources of finance other than existing distressed banks, such as new domestic banks, existing foreign banks, or securities markets.

beta. The residual beta is associated with 3-year forward P/E ratios, and thus with long-term earnings growth expectations and/or the cost of equity capital. We find that short-term earnings betas of vulnerable firms are *not* higher than those of non-vulnerable firms. Thus, the higher sovereign risk beta of vulnerable firms is fully attributable to higher sovereign risk sensitivity of 3-year forward P/E ratios. In other words, it is the long-term earnings growth expectations and/or the cost of equity capital of vulnerable firms that are more sensitive to sovereign risk.

Finally, we use a structural valuation model to examine the sovereign risk sensitivity of 3-year forward P/E ratios quantitatively. The purpose is to better understand the economic as opposed to statistical significance of such association. Model estimation indicates that stocks are priced as if the full effect of sovereign default on firms in the lowest tercile of asset tangibility amounts to a decrease in long-term earnings growth and/or an increase in cost of equity capital adding up to 1.58% per year. That is, while P/E ratio regressions show that the stock market *prices in* long-term effects of sovereign risk through credit disruption, the structural model estimation indicates that such effects are economically large.

1.1 Related literature and contribution

Recent research provides evidence supporting a financial intermediation channel for the cost of sovereign default. Gennaioli et al. (2014a) show that, following sovereign default, the decline in private credit is larger in those countries where the banking system holds more public bonds. In a follow-up paper, Gennaioli et al. (2014b) use bank-level data for a broad sample of countries and find that banks that own more domestic government bonds before sovereign default significantly contract their lending after default.

Many recent papers study bank lending during the European sovereign crisis and find effects consistent with the new theories of sovereign default. Popov and Van Horen (2014) and De Marco (2014) find that, controlling for credit demand, banks with greater holdings of risky sovereign debt tightened credit supply by more than less exposed banks. Bofondi et al. (2013) document that, compared to non-Italian banks operating in Italy, Italian banks tightened credit supply to non-financial Italian firms by lending less and at higher interest rates. Moreover, they find that the sovereign crisis indeed led to an aggregate contraction of credit, as Italian firms were not able to fully substitute credit from domestic banks with credit from non-Italian banks. Using survey data on small firms' access to bank credit, Ferrando et al. (2014) document a larger contraction in credit supply for firms in the GIIPS countries relative to firms in other Eurozone countries. They also find that the announcement of the ECB's Outright Monetary Transactions Program partially restored credit to small GIIPS firms.[7]

In an important paper, Acharya et al. (2014) take the evidence one step further. Using a matched sample of firms with syndicate loan information from Dealscan and accounting information from Amadeus, they first show that non-financial firms with higher exposure to banks holding more risky sovereign debt became more financially constrained during the crisis. Importantly, they also show that such high-exposure firms have significantly lower employment growth, capital expenditures, and sales growth rates in 2010 and 2011. That is, Acharya et al. (2014) link sovereign risk-induced credit crunch to outcomes in the real economy over a two-year period.

[7] Related research studies the association between sovereign risk and financial firms. Acharya and Steffen (2015) show that Eurozone banks with larger holdings of risky sovereign debt are more affected by widening sovereign spreads. Acharya, Drechsler and Schnabl (2014) and Alter and Beyer (2014) examine the feedback loop between sovereign and bank credit risk.

Our findings complement recent research and provide additional support to the new theories of sovereign default. Our approach, focusing on differential stock price sensitivity to sovereign spreads changes, is novel and provides evidence from a different perspective. Because our sample is comprised of publicly traded firms with median asset value of 1 billion euros, we show the financial intermediation costs of sovereign default are expected to extend beyond small or private firms. Because we find that stocks are priced as if the costs of financial intermediation disruption operate beyond the short-term (i.e., nearest three years), our results indicate that the frictions underlying the new theories of sovereign default can potentially have lasting consequences.

2 Data

We test whether stocks more vulnerable to financial intermediation disruption have greater sensitivity to changes in sovereign spreads. Our independent variables are stock returns. Our dependent variables are changes in sovereign spreads, and such changes interacted with an *ex ante* proxy for credit disruption vulnerability. We study Eurozone stocks from July 2005 to December 2013, a period that includes the Eurozone sovereign crisis. Data are monthly.

Stock returns

We obtain monthly firm-level, Euro denominated stock market data for Eurozone stocks on Datastream. We focus on non-financial stocks, as defined by ICB industry classifications. Stocks identified as cross-listings are deleted. In order to highlight the economic relevance of our results, we drop micro-caps from the sample, identified as the smallest stocks whose cumulative market capitalization add up to 1% of the total Eurozone market capitalization at the end of June each year. Finally, we drop stale Datastream observations, as evidenced by three identical consecutive values for total return index and market capitalization. On average, our sample has 892 stocks each month. Ours is an unbalanced panel: data encompass 1375 unique non-financial stocks, but only 558 of these are present throughout the entire sample period.

In addition to stock-level returns, we obtain monthly European four-factor return data from Ken French's website. Ken French's factor returns are U.S. dollar denominated. We use monthly dollar-euro exchange rate data from Datastream to express factor returns in Euros.

Sovereign spreads

We obtain end-of-month 10-year CDS spreads for Eurozone countries from Markit. All contracts are Euro-denominated and have the CR restructuring clause.[8] Figure 1 plots sovereign CDS spreads over time from 2004 to 2013. Note that spreads diverge strongly starting at the end of 2007.

FIGURE 1

To ensure that stock returns and CDS spread changes are in comparable units (i.e., returns), we compute normalized changes in spreads as follows:

$$\Delta Spread_t = \frac{Spread_t - Spread_{t-1}}{r_{F,t-1} + Spread_{t-1}} \left(1 - \frac{1}{(1 + r_{F,t-1} + Spread_{t-1})^{10}} \right)$$

where r_F is the 10-year yield on the German Bund minus the 10-year German CDS spread. Thus, our $\Delta Spread$ variable expresses, approximately, a monthly sovereign

[8] Sovereign CDS data for Greece end in February 2012. Sovereign CDS data for the Netherlands start in October 2005.

spread change as a return on a short position on a 10-year floating-rate sovereign bond (Duffie, 1999). The normalization is useful because, from an economic (e.g. bondholder) perspective, a CDS spread change from 100 to 200 basis points is economically different from a CDS spread change from 2000 to 2100 basis points, or from a CDS spread change from 1 to 2 basis points. Nonetheless, we verify that our results are robust to using raw changes in sovereign spreads.

Vulnerability to financial intermediation disruption

Our main proxy for vulnerability to financial intermediation disruption is asset tangibility. Firms with less tangible assets have less pledgeable collateral, and as such are more likely to be affected by a negative shock to aggregate credit supply (Holmstrom and Tirole 1997, Almeida and Campello 2007). Following the literature, we measure asset tangibility each calendar year using Property, Plant, and Equipment (PPE) divided by Total Assets. Accounting data are from Worldscope. To accommodate a possibly non-linear relationship between tangibility and vulnerability, and to guard against outliers, we use a dummy variable specification. Each month we rank firms based on asset tangibility. We then create the dummy variable *Low Tangibility*, flagging firms in the lowest tercile of asset tangibility each month.[9]

In robustness checks, we examine two additional proxies for vulnerability to financial intermediation disruption. Young firms have had less time to develop relationships with lenders, which makes them more vulnerable to negative shocks to total credit supply (e.g., Petersen and Rajan 1994, Chava and Purnanandam 2012). We measure age based on incorporation date using data from OSIRIS. The dummy

[9] In Appendix A we show that asset tangibility is a valid proxy to vulnerability to financial intermediation disruption in our particular setting. Specifically, we test whether *Low Tangibility* firms actually display signs of being credit distressed during periods over which sovereign spreads increase. We find that, relative to non-vulnerable firms, *Low Tangibility* experience a larger contraction in bank loans in periods in which sovereign spread increase.

variable *Young* flags firms incorporated after 2001.

Firms in economic sectors with intrinsically larger dependence on external finance to fund capital expenditures are also likely to be more strongly affected by aggregate credit disruption (Rajan and Zingales 1998). The dummy variable *High EF Dependence* flags firms in the highest tercile of the Rajan-Zingales external finance dependence metric. We use the Rajan-Zingales metric by ISIC sector from Claessens and Laeven (2003), generously provided to us by Luc Laeven, and the firm's ISIC classification from OSIRIS.[10]

Table 1 has summary statistics. In total, we have 91027 firm-month observations. France, Germany and Italy are the countries contributing the largest number of observations to the sample. The average excess stock return across all countries in the sample is 0.36% per month, with a standard deviation of 11%. Average stock returns are much higher in Germany (0.73% per month) than in Greece and Italy (-0.57% and 0.00% per month). The average (normalized) change in sovereign spread $\Delta Spread$ ranges from 0.01% per month in Finland to 1.76% per month in Greece. The standard deviation of $\Delta Spread$ across the entire sample is 3.26%, indicating there is considerable time-series and cross-sectional variability. Such variability is important for the empirical identification of sovereign risk betas.

Firms in our sample are large. The average of Total Assets (winsorized at 1%) is 7141 million euros. The median is 991 million euros. Across the entire sample, the average of the *Low Tangibility* dummy variable is equal to 0.334. All Eurozone countries contain both *Low Tangibility*=1 and *Low Tangibility*=0 observations. France

[10] There are firms in our (Datastream/Worldscope) sample for which we could not find an OSIRIS match. We cannot determine firm age for these firms, and hence they do not enter the definition of the *Young* variable. In computing the *High EF Dependence* metric, we assign firms without an ISIC code the average Rajan-Zingales metric for firms in the same Level 3 sector according to the Industry Classification Benchmark on Datastream.

is the country with the highest proportion of observations with *Low Tangibility*=1 (0.424). Greece is the country with the lowest proportion of *Low Tangibility*=0 observations (0.107).

TABLE 1

3 Main regressions

We test our main hypothesis using two empirical specifications: stock-level regressions and portfolio-level regressions using long-short portfolios.

Stock-level results

The starting point for return regressions is a four-factor model in which individual stock returns are driven by exposure to European wide market, size, value, and momentum factors (MKT, SMB, HMB, and WML), plus residual returns. We add an additional factor to the standard ones. Individual stocks (indexed by i) are exposed to an additional dimension of systematic risk, namely sovereign risk, which varies at the country level (indexed by C). We capture sovereign risk exposure by sensitivity to sovereign spread changes, *ΔSpread.*

$$r_{i,t} = \alpha + \beta\ \Delta Spread_{C,t} + \gamma_1\ MKT_t + \gamma_2\ SMB_t + \gamma_3\ HML_t + \gamma_4\ WML_t + \varepsilon_{i,t}$$

The new theories of sovereign default predict that non-financial stocks more vulnerable to financial intermediation disruption have higher sensitivity to sovereign risk, i.e., a higher beta with respect to *ΔSpread.* To test this prediction, we interact *ΔSpread* with the dummy variable *Low Tangibility.* To allow for the possibility that low asset-tangibility firms also have higher exposure to the traditional four factors, we also interact each of the four standard factors with *Low Tangibility*. If the coefficient on the interaction term $\Delta Spread \times$ *Low Tangibility* is negative and statistically significant, then our main hypothesis is confirmed. Note that we are essentially identifying cross-firm differences within a country. This is because the interaction term $\Delta Spread \times$ *Low Tangibility* captures differences across *Low Tangibility*=1 and *Low Tangibility*=0 firms for a given $\Delta Spread$, and the latter is defined at the country-level.

Table 2 contains results of our stock-level panel regressions. Because errors are likely to be cross-sectionally correlated within a country at a given point in time, we display t-statistics based on standard errors that are clustered at the country-time dimension.[11] Table 2, Column (1) has a preliminary result. We examine the contemporaneous correlation between non-financial stocks excess returns and sovereign spread changes. We find that stock returns are negatively correlated with sovereign spread changes, after controlling for four factor returns. On average, an increase in sovereign spreads resulting in a 1% change in $\Delta Spread$ is associated with a 0.135% decrease in stock returns. This sensitivity is statistically significant at the 1% level.

TABLE 2

Column (2) has the paper's main result. We find that *Low Tangibility* firms have higher sensitivity to sovereign risk. Compared to Column (1), we add interaction terms of our dummy variable *Low Tangibility* with $\Delta Spread$ and each of the four factors. We find that the interaction coefficient $\Delta Spread \times$ *Low Tangibility* is equal

[11] Our conclusions remain unchanged with White standard errors.

to -0.104. This coefficient is statistically significant at the 1% level. In the same regression, we find that the coefficient on *ΔSpread* is -0.120. Therefore, the sovereign risk beta of *Low Tangibility* firms (-0.120-0.104) is nearly double the sovereign risk beta of the other firms, and the difference is statistically significant at the 1% level. As the new sovereign default theories predict, firms that are more vulnerable to financial intermediation disruption are more sensitive to changes in sovereign spreads.

Columns (3) to (5) of Table 2 contain results of robustness checks. In Column (3) we add industry fixed effects, and find that our conclusions are unchanged. In Column (4) we restrict the sample period to the one most closely associated with the European sovereign debt crisis. The period starts in January 2010 instead of July 2005. In Column (5) we restrict the sample to stocks from Greece, Ireland, Italy, Portugal, and Spain (GIIPS). The results in Columns (3) to (5) are qualitatively and quantitatively similar to those in Column (2). In sum, consistent with the new sovereign default theories, stock-level panel regressions show that firms that are expected to suffer more from a financial intermediation disruption, as measured by *Low Tangibility*, display more sensitivity to sovereign risk.[12]

Portfolio results

Table 3 contains results of our portfolio-level regressions. Within each Eurozone country, we form portfolios long stocks with *Low Tangibility* equal to 1 and short the remaining stocks with *Low Tangibility* equal to zero. Both equally-weighted and value-weighted long-short portfolios are formed. We regress long-short portfolio returns onto contemporaneous changes in sovereign spreads and four factor returns.

[12]In untabulated regressions we add country and time fixed effects. Our conclusions remain unchanged. The resulting *Low Tangibility* $\times$ Δ*Spread* coefficients are -0.104 (t-stat 2.71), -0.105 (t-stat -2.99), and -0.106 (t-stat 3.01) respectively when we add country, time, and both country and time fixed effects. Such coefficient estimates are very close to those in Table 3.

The 11 Eurozone long-short portfolios are stacked so that we explore not only time series but also cross-country variation in *ΔSpread*. We report three types of standard errors: White, Driscoll-Kraay, and clustered at the time level.[13]

TABLE 3

Columns (1) to (3) report results of regressing equally-weighted *Low Tangibility* long-short portfolio returns on changes in sovereign spreads. Columns (2) and (3) control for the four factors, i.e., they allow for the possibility that long and short portfolios based on *Low Tangibility* have different exposures to MKT, HML, SMB, and WML. In Column (3) we add interactions of each of the four factors with country dummies, to allow for different four factor betas for each of the long-short country portfolios.

Results in Columns (1) to (3) indicate that *Low Tangibility* portfolios stocks have higher exposure to sovereign risk than non-*Low Tangibility* portfolios. The coefficient on *ΔSpread* is negative in all these columns. The coefficients are statistically significant at the 5% level using all types of standard errors. Columns (4) to (6) have results of using value-weighted instead of equally-weighted long-short portfolios. The coefficient on *ΔSpread* is negative all three columns, and statistically significant at 5% using all types of standard errors.

The portfolio results agree with stock-level results. Stocks more vulnerable to financial intermediation disruption have larger sovereign risk betas. This supports the new theories of sovereign default by Gennaioli et al. (2014) and others.

[13] We report Driscoll-Kraay standard errors with 4 lags. Results are robust to using either 1, 2, or 3 lags.

4 Additional regressions

4.1 Alternative vulnerability proxies

In this section we proxy for vulnerability to financial intermediation disruption using the Rajan-Zingales external finance metric and firm age instead of asset tangibility. As in Table 2, we regress individual stock returns onto *ΔSpread* and the four factors, including interactions with *High EF Dependence* or *Young.* Results in Table 4 show that our conclusions are robust to using these alternative proxies. As the new theories of sovereign default predict, firms that are *ex ante* more vulnerable to financial intermediation disruption are more sensitive to changes in sovereign spreads.

TABLE 4

Columns (1) shows that *High EF Dependence* firms have higher sovereign risk sensitivity. The coefficient on the *ΔSpread* × *High EF Dependence* interaction term is -0.086 and statistically significant at the 1% level. Therefore, *High EF Dependence* firms have sovereign risk betas (-0.118-0.086) that are 72% larger than those of other firms. Column (3) shows that *Young* firms also have higher sensitivity to sovereign risk. The coefficient on the *ΔSpread* × *Young* interaction term is -0.159, and significant at the 10% level. Therefore, the sovereign risk beta of *Young* firms (-0.122-0.159) is more than double the sovereign risk of the other firms. Columns (2) and (4) show that results are robust to adding industry fixed effects.

4.2 Announcement returns

Our stock-level and portfolio-level regressions show that stocks more vulnerable to financial intermediation disruption display higher sensitivity to changes in sovereign spreads, after controlling for exposure to standard stock return factors. This is consistent with the new theories of sovereign default, in which sovereign default affects the economy by upsetting domestic financial intermediation. One may worry, however, that vulnerable stocks are more sensitive to sovereign spread changes because of an omitted variable that affects spreads and systematically affects vulnerable stocks more than non-vulnerable ones.

In order to more directly link stock returns to news about sovereign default, we focus on short-term returns immediately following three policy announcements. These announcements had large effects on financial markets. If firms more vulnerable to financial intermediation disruption display higher sensitivity to sovereign spread changes over these short windows, then concerns about an omitted variable are mitigated.

Our first event is the May 8, 2010 announcement of the creation of the European Financial Stability Facility (EFSF). That EFSF was created to alleviate the European sovereign-debt crisis. The Fund would provide long term lending to European sovereigns at below market rates so that countries could roll over their debts. The second event is the July 26, 2012 speech by Mario Draghi, the president of the European Central Bank (ECB), at the Global Investment Conference in London. Amidst market concerns that countries would countries default on their debts and break up from the Euro, Draghi stated "*Within our mandate, the ECB is ready to do whatever it takes to preserve the euro. And believe me, it will be enough.*" The third event is the announcement of the creation of the ECB's Outright Monetary Transactions (OMT) program on September 6, 2012. The OMT program allowed the ECB to purchase European sovereign bonds directly on the secondary market.

TABLE 5

Table 5, Panel A shows all three events elicited strong, positive reaction in both sovereign spread and stock markets. We report average two-day returns following each of the three events. On average across the Eurozone countries, our *ΔSpread* variable (sovereign spread change in unit of returns) is -0.0418 following the EFSF creation, -0.0174 following Draghi's speech, and -0.0167 following the OMT creation.[14] These changes are more than three standard deviations from zero for all three events. On average across all Eurozone stocks, stock prices increased 0.0472, 0.0381, and 0.0231 following the first, second, and third events, respectively. These first- and second-event average returns are more than two standard deviations from zero.

Table 5, Panel B shows that, for a given sovereign spread change, stock returns of *Low Tangibility* firms react more strongly to the three events. We stack the three events and run regressions of announcement stock returns onto *ΔSpread* and *ΔSpread* interacted with *Low Tangibility.* We report t-statistics based on standard errors clustered at the country-level.

Column (1) has a preliminary result. It shows that, following the announcements, stock returns increased more strongly in countries in which sovereign spreads decreased more strongly. Column (2) shows that, for a given change in *ΔSpread*, *Low Tangibility* stocks reacted more strongly to the announcements.

[14] There are no Greek stocks in the second and third events because these events occur after Greek sovereign CDS data becomes unavailable.

Columns (3), our preferred specification, shows that *Low Tangibility* stocks are more sensitive to sovereign risk after controlling for exposure to the four factors and industry membership. Stock-level betas with respect to the four factors, computed using monthly data from June/2005 to December/2013, and industry fixed effects are added as controls. As in Table 2, the sensitivity of stock returns to sovereign risk is about double for *Low Tangibility* stocks compared to other stocks (-0.084-0.107 versus -0.084). The difference is statistically significant at the 5% level. Column (4) shows that our conclusions are unchanged when restricting the sample to Greek, Irish, Italian, Portuguese and Spanish stocks. We conclude that vulnerable stocks have higher exposure to sovereign risk not only in general, but also in short-windows following policy announcements pertaining to intervention in sovereign debt markets.

4.3 Sovereign beta decomposition

Our baseline results in Table 2 show that stock returns of firms more vulnerable to financial intermediation disruption have higher sovereign risk sensitivity. In this section we further examine such sensitivity by decomposing sovereign risk betas into an immediate and a long-term component. The immediate component is related to earnings expectations over the nearest three years. The long-term component is associated with 3-year forward P/E ratios, and hence with long-term earnings growth and/or the cost of equity capital. We find that the higher sovereign risk sensitivity of vulnerable firms derives exclusively from the long-term component.

We first decompose stock returns excluding dividends into changes in short-term (up to three years) earnings expectations (E) and changes in P/E ratios computed using such earnings expectations (i.e., changes in 3-year forward P/E ratios). Then we separately regress each change onto contemporaneous changes in sovereign spreads.

The following notation helps. Denote E_t as earnings-per-share (EPS) expectations at time t. We can re-write stock returns (excluding dividends) as:

$$\begin{aligned}\frac{P_{t+1}}{P_t} &= \frac{P_{t+1}}{E_{t+1}}\frac{E_{t+1}}{E_t}\frac{E_t}{P_t} \\ &= \frac{\frac{P_{t+1}}{E_{t+1}}}{\frac{P_t}{E_t}}\frac{E_{t+1}}{E_t}\end{aligned}$$

Taking logs on both sides, assuming earnings expectations are positive, yields:

$$\log\left(\frac{P_{t+1}}{P_t}\right) = \left[\log\left(\frac{P_{t+1}}{E_{t+1}}\right) - \log\left(\frac{P_t}{E_t}\right)\right] + \left[\log\left(E_{t+1}\right) - \log\left(E_t\right)\right] \tag{1}$$

Equation 1 shows that log price changes can be decomposed onto changes in log P/E ratios and changes in log earnings expectations. Thus, if a firm's stock return is sensitive to sovereign spread changes, it must be because its earnings expectations, or its P/E ratio, or both, are sensitive to spread changes.

We use mean EPS expectations from I/B/E/S for up to three years. Specifically, we average EPS forecasts for the current fiscal year, the next fiscal year, and the fiscal year after that. Because the decomposition in Equation 1 requires positive earnings, we delete observations with negative earnings.[15] These three-year average EPS expectations are matched to Datastream price data as of the same day. Because earnings expectations from I/B/E/S may be infrequently updated by sell-side analysts, our baseline regressions have quarterly (March, June, September, and December) rather than monthly observations.[16] Conclusions are unchanged when we use monthly data.

[15] The deletion of negative earnings forecasts reduces the sample (firm-quarters) by 3%. Later we show that the deletion does not affect our conclusions.

[16] I/B/E/S mean forecasts are computed at the third Thursday of each month. I/B/E/S mean forecasts use forecasts made over the previous 105 days. Older forecasts are considered stale.

Table 6 (Panel A) contains our results. The first two columns report regressions of changes in log prices onto *ΔSpread*. Note these columns simply reproduce Table 2 baseline results with the following four changes. We use changes in log prices instead of stock returns on the left-hand side, use quarterly instead of monthly data, focus on the subset of firms for which we have I/B/E/S data, and exclude four factor returns.[17] Column (2) shows that the interaction term *ΔSpread* × *Low Tangibility* is negative and statistically significant. That is, consistent with Table 2 baseline results, firms more vulnerable to financial intermediation disruption display higher sensitivity to sovereign risk.

TABLE 6

Columns (3) and (4) of Table 6 (Panel A) report regressions of log P/E ratios changes onto *ΔSpread*. Columns (5) to (6) have regressions of changes in log short-term earnings expectations onto *ΔSpread*. Because changes in log P/E ratios and changes in log earnings expectations add up to changes in log prices (Equation 1), the regression coefficients also add up. That is, the coefficients on Column (1) are algebraically identical to the sum of the corresponding coefficients on Columns (3) and (5). Analogously, Column (2) coefficients are the sum of coefficients in Columns (4) and (6). Note that the coefficients on *ΔSpread* are negative and statistically significant in both Columns (3) and Column (5). That is, as sovereign spreads increase, short-term earnings expectations are revised downward and stocks become "cheaper" in terms of 3-year forward P/E ratios.

Importantly, however, the interaction term *ΔSpread* × *Low Tangibility* is negative and statistically significant in Column (4) but not in Column (6). Therefore, relative

[17] Including four factor returns and corresponding interactions with *Low Tangibility* does not change our conclusions from this section.

to other stocks, short-term earnings expectations of credit vulnerable firms do *not* display higher sensitivity to sovereign spread changes. These firms' stock returns display higher sovereign risk betas entirely because their forward P/E ratios are relatively more sensitive to sovereign spread changes. This indicates that the higher sovereign risk beta of credit vulnerable firms reflects higher sensitivity of market expectations about long-term (beyond three years) earnings growth and/or the cost of equity capital.

Panel B of Table 6 shows that our dropping of negative earnings forecasts does not influence the key conclusion drawn from Panel A of Table 6. The left-hand side variable of Panel B table is quarterly change in EPS expectations ($\mathrm{E}_{t+1}-\mathrm{E}_t$) *without* dropping negative earnings forecasts. These changes are scaled by stock prices in three different ways, all of them deliberately shutting down time series variations due to price changes. In Columns (1) and (2) we scale by the first observed stock price in the matched sample. In Columns (3) to (4) we scale by the last observed stock price in the matched sample. In Columns (5) and (6) we scale by the average stock price in the matched sample.

Regardless of how earnings expectations changes are scaled, Panel B of Table 6 shows that the interaction term *ΔSpread* × *Low Tangibility* is economically small and statistically insignificant in all columns. That is, when sovereign spreads increase, credit vulnerable firms do not display stronger reduction in earnings expectations relative to other firms. Thus, the higher sovereign risk beta of vulnerable firms does not reflect short-term (below three years) earnings expectations. Rather, it is due to these firms 3-year forward P/E ratios being relatively more sensitive to sovereign risk, which indicates market concerns about longer-term effects of sovereign risk.

5 Using a structural valuation model

In the previous section we find that firms vulnerable to financial intermediation disruption have higher sensitivity to sovereign risk because their 3-year forward P/E ratios are more sensitive to sovereign spread changes. In this section we use a simple valuation model to gauge the economic significance of such sensitivity. The model allows us to estimate the full effect of sovereign default on vulnerable firms that is implicit in market prices.

The model, adapted from Andrade (2009), assumes that sovereign default reduces long-term earnings growth and increases the cost of equity capital for some stocks. From the relationship between sovereign spreads and the P/E ratio differential across stocks affected and not affected by sovereign default, the model allows us to estimate the effect of default implicit in market prices. Model estimation shows that sovereign default is expected to have an economically large long-term effect on firms vulnerable to financial intermediation disruption. The model is briefly described below.

Let a country have foreign debt requiring a continuous and constant payment flow $c > 0$. The total foreign debt service is composed of coupon and principal payments. The country continuously retires a fraction $\frac{1}{L}$ of its total debt, replacing it by newly issued debt with the same principal value and coupon rate. Leland (1994, 1998) shows that this framework allows for constant total debt service and finite average debt maturity L, while total payments are exponentially declining for each debt vintage.

The country can default on its foreign debt. After default, the debt payment flow is reduced to $0 < \bar{c} < c$. That is, the recovery rate on defaulted sovereign debt is $R = \frac{\bar{c}}{c}$. After default, total payments are also composed of coupon and principal payments retired at a rate $\frac{1}{L}$, so that average debt maturity after default remains L. Let the average yield spread on outstanding sovereign debt be S_t and the risk-free rate be r. The following proposition shows how the risk-neutral probability of default implied by sovereign debt values, denoted Q_t, relates to S_t, R, r, and L.

Proposition 1 *The risk-neutral sovereign default probability is equal to*

$$Q_t = \frac{S_t}{(1-R)\left(S_t + r + \frac{1}{L}\right)} \tag{2}$$

Proof. See Appendix C.■

The new theories of sovereign debt propose that sovereign default is costly because it disrupts financial intermediation. Stocks vulnerable to such disruption will be directly affected by sovereign default, while non-vulnerable stocks will not. The model focuses on such difference across stocks within a country.

Let the earnings flow of a non-vulnerable stock follow a Geometric Brownian Motion (GBM) with trend μ_x, volatility of earnings growth σ_x, and correlation ρ_x with the global pricing kernel. Assume the global pricing kernel follows a GBM with trend equal to (minus) the international risk-free rate r and volatility equal to (minus) the global price of risk λ. Therefore, the cost of equity capital (or discount rate) of the non-vulnerable stock is equal to $d = r + \lambda\rho_x\sigma_x$ and its earnings yield (inverse P/E ratio) is $(\frac{E}{P})^{non-vuln.} = d - \mu_x$.

Now consider a vulnerable stock that is otherwise *comparable* to the non-vulnerable stock above. That is, though vulnerable and non-vulnerable stocks are driven by separate GBM processes, such processes have identical parameters. Unlike the non-vulnerable stock, vulnerable stock is directly affected by sovereign default. After default, the vulnerable stock experiences a higher cost of equity capital ($\overline{d} > d$) and a lower earnings growth rate ($\overline{\mu_x} < \mu_x$). Define the cost of default K_0 as the sum of the increase in the cost of equity capital and the decrease in earnings growth rate following sovereign default:

$$K_0 \equiv \left(\overline{d} - d\right) + \left(\mu_x - \overline{\mu_x}\right) \tag{3}$$

The vulnerable stock trades at a discount relative to the non-vulnerable one. This discount arises because stock prices reflect the possibility of a negative regime change following sovereign default. If such regime change tends to happen in bad economic times, there will also be a systematic risk premium associated to the discount, above and beyond the likelihood of the regime change alone.

Define the Value Discount as the relative P/E differential between the vulnerable and the non-vulnerable stock:

$$VD_t \equiv \frac{\left(\frac{P}{E}\right)^{non-vuln.} - \left(\frac{P}{E}\right)_t^{vulnerable}}{\left(\frac{P}{E}\right)^{non-vuln.}} \tag{4}$$

The following proposition shows how to link the Value Discount to the risk-neutral probability of sovereign default.

Proposition 2 *Let the country default the first time a Geometric Brownian Motion Y_t with trend μ_y and volatility σ_y hits an exogenous lower barrier $\overline{Y}$.*[18] *Let the correlations of Y_t with the earnings flow of the stock and with the global pricing kernel be ρ_{xy} and ρ_y, respectively. Then the Value Discount is equal to*

$$VD_t = \frac{K_0}{(\frac{E}{P})^{non-vuln.} + K_0} Q_t^{K_1} \tag{5}$$

where

$$K_1 = \frac{\frac{1}{2} - \frac{\mu_y - \lambda\sigma_y\rho_y}{\sigma_y^2} - \frac{\rho_{xy}\sigma_x}{\sigma_y} - \sqrt{\left[\frac{1}{2} - \frac{\mu_y - \lambda\sigma_y\rho_y}{\sigma_y^2} - \frac{\rho_{xy}\sigma_x}{\sigma_y}\right]^2 + \frac{2}{\sigma_y^2}(r + \lambda\sigma_x\rho_x - \mu_x)}}{\frac{1}{2} - \frac{\mu_y - \lambda\sigma_y\rho_y}{\sigma_y^2} - \sqrt{\left(\frac{1}{2} - \frac{\mu_y - \lambda\sigma_y\rho_y}{\sigma_y^2}\right)^2 + \frac{2}{\sigma_y^2} r}} > 0$$

Proof. See Appendix C.■

Appendix C refers to Andrade (2009) for a formal derivation of Equation (5). It also contains an heuristic derivation that helps to clarify the equation's origin. Equation (5) shows that two parameters, K_0 and K_1, govern the link between the Value Discount and the risk-neutral probability of sovereign default. The parameter K_0 determines the cost of sovereign default, as defined in Equation (3). The parameter K_1 governs the translation from a bond-based to a stock-based risk-neutral default probability.[19]

[18] Andrade (2009) endogenizes $\overline{Y}$ by defining the country's net endowment as $Y_t - c$ and assuming that default causes a decrease in the trend and an increase in the volatility of Y_t. In this case, $\overline{Y}$ becomes an optimal stopping time when the country seeks to maximize the present value of its net endowment. Sovereign default is also more likely to take place when the country is facing negative shocks in the general equilibrium models of Arellano (2008) and Borri and Verdelhan (2012), and elsewhere in the sovereign debt literature. Because the stochastic process Y_t can be correlated with the pricing kernel, the risk-neutral probability of default Q_t can diverge from its physical counterpart, as in Borri and Verdelhan (2012).

[19] If the price of risk is zero ($\lambda = 0$), and earnings have zero drift ($\mu_x = 0$) and are uncorrelated with the Y_t process determining sovereign default ($\rho_{xy} = 0$ and), then $K_1 = 1$.

The parameters K_0 and K_1 can be estimated from the empirical relationship between the Value Discount and sovereign spreads. In order to estimate them by maximum likelihood, we allow for additive specification/measurement errors ε_t. We assume that such errors are i.i.d. and normally distributed, but do not impose that they are mean zero.[20] And, to take the model to the data, we allow for time variation in the parameters $(\frac{E}{P})^{non-vuln.}$, R, L, and r. Thus, our estimation equation is:

$$VD_t = \frac{K_0}{\left(\frac{E}{P}\right)_t^{non-vuln.} + K_0} Q_t^{K_1} + \varepsilon_t \quad \text{, with } Q_t = \frac{S_t}{(1 - R_t)\left(S_t + r_t + \frac{1}{L_t}\right)} \ . \tag{6}$$

5.1 Model estimation

We need to define sets of comparable stocks in order to estimate the model. We postulate that stocks in the same country and industry are comparable. The industry grouping follows Bekaert et al. (2007, 2011, 2013), who propose that, under economic and financial integration, stocks in the same industry should have the same growth opportunities and discount rates. Within each country-industry set of comparable stocks, vulnerable stocks have *Low Tangibility*=1 and non-vulnerable stocks have *Low Tangibility*=0.

We obtain quarterly observations on VD_t and $(\frac{E}{P})_t^{non-vuln.}$ for country-industry pairs as follows. We start from quarterly stock-level 3-year forward P/E ratios used in Panel A of Table 6. For *Low Tangibility*=1 and *Low Tangibility*=0 stocks separately, we aggregate data to the country-industry level by value-weighting stock-level earnings yields. We value-weight using each stock's market capitalization, following Bekaert et al. (2011). Thus, we obtain quarterly $(\frac{E}{P})_t^{vulnerable}$ and $(\frac{E}{P})_t^{non-vuln.}$ for each industry-country pair.

[20] As Liu, Whited, and Zhang (2009, p.1110) point out, measurement and specification errors, unlike forecast errors, do not necessarily have mean equal to zero.

While the model assumes that earnings yields of comparable stocks are identical when sovereign risk is negligible, it is possible that earnings yields are affected by characteristics other than country and industry membership. This would lead us to attribute to sovereign risk differences in $\left(\frac{E}{P}\right)_t^{vulnerable}$ and $\left(\frac{E}{P}\right)_t^{non-vuln.}$ that are in fact spurious. To address this possibility, we add to all $\left(\frac{E}{P}\right)_t^{non-vuln.}$ the average difference between $\left(\frac{E}{P}\right)_t^{vulnerable}$ and $\left(\frac{E}{P}\right)_t^{non-vuln.}$ during the first 10 observations in our sample period (June 2005 to September 2007). Because sovereign risk was negligible during that period as indicated by CDS spreads, this adjustment guarantees that residual differences in earnings yields across firms (net of country and industry effects) are not spuriously attributed to sovereign risk. Finally, we compute VD_t for each country-industry pair as in the definition in Equation (4).

Note that not all country-industry pairs are populated. Some countries have no stocks in a given industry. Other countries do not have both *Low Tangibility*=1 and *Low Tangibility*=0 stocks in the same industry, so a Value Discount cannot be defined. Some industry-country pairs are very thinly populated, with for example only one *Low Tangibility* stock. In our baseline estimation we require at least three *Low Tangibility*=1 stocks and at least three *Low Tangibility*=0 stocks. Appendix Table A.2 has information about the industry-country pairs available to us.

We obtain quarterly Q_t observations for each country as follows. The sovereign spread S_t is the same 10-year sovereign CDS spread used in Panel A of Table 6. The recovery rate R_t comes from Markit's survey of CDS dealers. Such recovery rates are used by dealers to mark-to-market their positions, and to unwind CDS trades. The risk-free rate r_t is the yield on the benchmark 10-year German Bund minus the 10-year German CDS spread. The maturity of sovereign debt L_t is equal to 10 years, because we observe 10-year CDS spreads. Equation (2) yields Q_t given S_t, R_t, r_t, and L_t .

Panel A of Table 7 has summary statistics. There are 725 country-industry-quarter observations for which there are at least three *Low Tangibility*=1 stocks and at least three *Low Tangibility*=0 stocks. The average earnings yield of non-vulnerable stocks $\left(\frac{E}{P}\right)_t^{non-vuln.}$ is 0.080, and the average Value Discount VD_t is 0.024. Note there is a lot of dispersion in VD_t, as indicated by a standard deviation of 0.356, while the dispersion in $\left(\frac{E}{P}\right)_t^{non-vuln.}$ is not that large compared to its average level. The average (bond-based) risk-neutral probability of default Q_t is 0.081, and its standard deviation is 0.094.[21]

TABLE 7

Results in Panel B show that the cost of sovereign default is expected to be large. There we report results of estimating model parameters K_0 and K_1 by maximum likelihood.[22] We find a cost of default K_0 equal to 1.58% per year in our baseline specification, statistically significant at the 1% level. That is, following sovereign default, vulnerable firms are expected to experience a large decrease in their long-run rate of earnings growth and/or a large increase in their cost of equity capital. The K_1 estimate is equal to 0.269, indicating that the relationship between VD_t and Q_t is strongly non-linear.

Panel B also shows that our basic conclusions are robust to alternative estimations. In the second and third columns we report results in which less stringent filters are applied to the data. In the second column we do not apply any filter and

[21] According to the structural model, a linear regression of VD_t onto $\left(\frac{E}{P}\right)_t^{non-vuln.}$ and Q_t is mis-specified. Nonetheless, we report results of such regression on Appendix Table A.3. Confirming model predictions, we find VD_t and Q_t are positively related. Such positive relationship is not surprising given results in Panel A of Table 6. That table already implies that when sovereign spreads increase, forward P/E ratios of vulnerable firms decrease relative to forward P/E ratios of non-vulnerable firms, that is, Value Discounts increase.

[22] Note that maximum likelihood is equivalent to non-linear least squares when errors are i.i.d. normally distributed. We, however, report standard errors that are clustered at the country level.

use country-industry observations potentially based on just one *Low Tangibility*=1 or just one *Low Tangibility*=0 stock. As a result, the number of country-industry quarterly observations nearly doubles, increases from 725 to 1472.[23] Model fit worsens, as the correlation between fitted and observed Value Discount drops from 0.475 to 0.291. However, the cost of default K_0 estimate changes little, decreasing from 1.58% per year to 1.34% per year. The parameter K_0 still is statistically significant at the 1% level. In the third column we require two instead of three stocks in each vulnerability status. Again, model fit worsens relative to the baseline, but the the estimate K_0 does not change much.

In the fourth column of Panel B we apply a more stringent filter, requiring at least four stocks in each vulnerability status. The number of quarterly observations drops from 725 to 576. Model fit improves marginally, but, again, our conclusions are not significantly affected. The cost of default parameter K_0 barely increases, changing from 1.58% to 1.61% per year. It still is statistically significant at the 1% level.

Finally, in the last column of Panel B we restrict the sample to the GIIPS. The sample size drops substantially, from 735 to 234. The cost of default K_0 estimate changes little, decreasing from 1.58% per year to 1.50% per year, and is statistically significant at the 5% level. Model fit improves a little, as the correlation between actual and fitted values increases from 0.48 to 0.54. From the alternative estimations, we conclude that the basic result of Panel B is robust: stock prices behave as if sovereign default has an economically large impact on firms that are vulnerable to disruption in financial intermediation.

[23] In this specification we drop three observations (out of the original 1475) in which the Value Discount is more 10 standard deviations from the cross-sectional average.

6 Conclusion

This paper provides novel support for theories of sovereign debt proposing that domestic financial intermediation disruption is the key cost of sovereign default (e.g., Gennaioli et al. 2014a). We find that firms more vulnerable to credit market disruption have higher stock return sensitivity to changes in sovereign credit spreads. Specifically, stocks with low asset tangibility have sovereign risk betas that are twice as large as other stocks. The higher sovereign risk sensitivity of vulnerable firms also obtains in short-window returns immediately following announcements pertaining to policy intervention in sovereign debt markets.

We show that the higher sovereign beta of vulnerable firms does *not* obtain because their short-term (less than three years) earnings expectations are especially sensitive to sovereign spreads. The higher sensitivity is fully due to higher sensitivity of forward P/E ratios, which depend on market expectations about long-term earnings growth rates and the cost of equity capital. Therefore, stocks are priced as if financial intermediation disruption due to sovereign default has long-term consequences.

We estimate a structural valuation model to quantitatively examine the empirical association between forward P/E ratios and sovereign spreads. Model estimation indicates that the long-term cost of sovereign default on vulnerable firms is expected to be economically large. Specifically, stocks are priced as if sovereign default leads to a decline in the rate of long-term earnings growth coupled with an increase in the cost of equity capital adding up to 1.58% per year. Future research studying whether and why the financial frictions that give rise to domestic credit disruption have long-lived consequences should provide additional insights on the economics of sovereign default.

Appendix A: Validating the vulnerability proxy

Following the literature, our proxy for vulnerability to financial intermediation disruption is asset tangibility. A natural question is whether this is a good proxy in our particular setting. To address this question we test whether Eurozone *Low Tangibility* firms actually display signs of being credit distressed during periods over which sovereign spreads increase. To that end we investigate changes in the amount of bank loans that firms have taken (scaled by total assets). Relative to non-vulnerable firms, credit vulnerable firms should experience a larger contraction in bank loans in periods of sovereign spread increases.

TABLE A.1

Table A.1 shows results. We run regressions with annual changes in bank loans (scaled by assets), winsorized at the 1% level, on *ΔSpread* and *ΔSpread* interacted with *Low Tangibility*. Data are from Eurozone firms in OSIRIS and range from 2005 to 2013. Column (1) has a preliminary result showing that, on average across firms, bank loans do contract when sovereign spreads increase. Column (2) shows that, for a given increase in sovereign spreads, *Low Tangibility* firms indeed experience a much larger reduction in bank loans. Specifically, the decline is 381% larger for *Low Tangibility* firms: (-0.016-0.045)/(-0.016). The difference is statistically significant at the 5% level. Column (3) shows conclusions are unchanged when industry fixed effects are included.

Appendix B: European stock market (dis)integration

Table 6 (Panel A) shows that European P/E ratios are sensitive to sovereign risk. This subsection provides an alternative lens to viewing such results. We show that the emergence of sovereign risk in Europe temporarily disrupted a multi-decade trend towards the equalization of intra-industry P/E ratios across countries. Such trend is first documented by Bekaert et al. (2011), who posit that the trend is due to financial and economic globalization leading to the worldwide equalization of both discount rates and growth opportunities. Our results are consistent with Bekaert et al. (2011), who find that political risk – which encompasses sovereign risk – can counteract globalization forces and consequently widen P/E ratio differentials.

Following Bekaert el al. (2011), we calculate a segmentation measure based on pairwise absolute differences of industry level earnings yields (inverse of P/E ratios) across countries. Differently from them, we use monthly earnings yields based on average earnings forecasts up to three years rather than annual yields based on the previous year historical earnings. We start with firm-level earnings yields each month. These are aggregated to the industry-level within each country using value-weighting. Then we compute the weighted-average absolute difference of earnings yields across industries for each country-pair, value-weighting industries like Bekaert et. al. (2011). Finally, we average across all country-pairs to find one cross-country segmentation measure each month.

Figure A.1 has the time series of segmentation calculated for European developed markets. For comparison, we also plot the segmentation calculated for non-European developed markets. Data are monthly from 1987 to 2013.

FIGURE A.1

Figure A.1 shows the long-term trend towards declining earnings yields differentials, confirming Bekaert et al. (2011). The figure also shows that valuation ratios sharply diverged in the Fall of 2008, with the price dislocations associated with the Global Financial Crisis (see Pasquariello 2014). Such dislocations where short-lived in the rest of the world, as earnings yield quickly re-converged and differentials end the sample near the historical low. However, in Europe the emergence of a sovereign debt crisis led to increasing segmentation parallel to the decreasing segmentation in the rest of the world. In the second half of 2012 the European crisis subsides, and European earnings yields differentials narrow again.

Appendix C: Structural model Proofs

Proposition 1.

Consider riskless debt of a given vintage, paying $c > 0$ at an exponentially declining rate $m = \frac{1}{L}$. The value of this debt is

$$B^{riskless} = E_t[\int_t^{\infty} e^{-r(s-t)} e^{-m(s-t)} c \; ds] = \frac{c}{r+m}.$$

Let the value of sovereign debt be B_t. By definition, the sovereign spread S_t is such that $B_t = \frac{c}{r+S_t+m}$. On the other hand, by definition of the risk-neutral default probability, we must have:

$$B_t = \frac{c}{r+m} - Q_t \left(1-R\right) \frac{c}{r+m}$$

Therefore, from the two equations for B_t we get:

$$\frac{c}{r+S_t+m} = \frac{c}{r+m} - Q_t \left(1-R\right) \frac{c}{r+m}$$

, which solving for Q_t and using $m = \frac{1}{L}$ yields equation (2) in the text:

$$Q_t = \frac{S_t}{\left(1-R\right)\left(S_t + r + \frac{1}{L}\right)}$$

Proposition 2 (Formal Derivation).

Take Equation (5) in Andrade (2009). Consider that the acronyms EM and DEV denote vulnerable and non-vulnerable stocks respectively. Then make $\eta = 0$, $r + \lambda\rho_x\overline{\sigma_x} = \ \overline{d}$, $\frac{\overline{c}}{c} = R$, and $\frac{\alpha}{\beta} = K_1$ to obtain:

$$\left(\frac{P}{E}\right)_t^{vulnerable} = \frac{1}{(r + \lambda\rho_x\sigma_x - \mu_x)\left(\overline{d} - \overline{\mu_x}\right)}\left(\overline{d} - \overline{\mu_x} - \left[\frac{S_t}{(1-R)(S_t + r + m)}\right]^{K_1}\left[\overline{d} - \overline{\mu_x} + \mu_x - \overline{\mu_x}\right]\right)$$

Now substitute $\left(\frac{E}{P}\right)^{non-vuln.} = d - \mu_x = r + \lambda\rho_x\sigma_x - \mu_x$and $K_0 = \left(\overline{d} - d\right) + (\mu_x - \overline{\mu_x})$ in the equation above to get:

$$\left(\frac{P}{E}\right)_t^{vulnerable} = \frac{1}{\left(\frac{E}{P}\right)^{non-vuln.}}\ \frac{\overline{d} - \overline{\mu_x} - \left[\frac{S_t}{(1-R)(S_t+r+m)}\right]^{K_1} K_0}{\left(\overline{d} - \overline{\mu_x}\right)}$$

Therefore,

$$\frac{\left(\frac{P}{E}\right)_t^{vulnerable}}{\left(\frac{P}{E}\right)^{non-vuln.}} = 1 - \left[\frac{S_t}{(1-R)(S_t + r + m)}\right]^{K_1}\frac{K_0}{\overline{d} - \overline{\mu_x}}$$

Now note that $\overline{d} - \overline{\mu_x} = \left(\frac{E}{P}\right)^{non-vuln.} + K_0$. Substituting this in the equation above gives:

$$\frac{\left(\frac{P}{E}\right)_t^{vulnerable}}{\left(\frac{P}{E}\right)^{non-vuln.}} = 1 - \left[\frac{S_t}{(1-R)(S_t + r + m)}\right]^{K_1}\frac{K_0}{\left(\frac{E}{P}\right)^{non-vuln.} + K_0}$$

Using the equation for Q_t in Proposition 1 and re-arranging terms in the equation above yields Equation (5) in the text:

$$VD_t \equiv 1 - \frac{\left(\frac{P}{E}\right)^{non-vuln.} - \left(\frac{P}{E}\right)_t^{vulnerable}}{\left(\frac{P}{E}\right)^{non-vuln.}} = \frac{K_0}{\left(\frac{E}{P}\right)^{non-vuln.} + K_0} Q_t^{K_1}.$$

Proposition 2 (Heuristic Derivation).

To aid intuition, let's start reviewing the role of the bond-based risk-neutral probability of default Q_t in bond pricing. Let the value of a risk-free debt be denoted by $B^{riskless}$. The value of risky sovereign debt is:

$$B_t = B^{riskless}\left[1 - (1 - R)\, Q_t\right]$$

Now consider an equivalent stock-based rather than bond-based equation. We will find the "stock-equivalent" of each of the objects B_t, $B^{riskless}$, $(1 - R)$, and Q_t in the bond-based equation.

First, on the left-hand side of the equation we have the price of the vulnerable stock in place of B_t. Normalizing by earnings, the left-hand side has $\left(\frac{P}{E}\right)_t^{vulnerable}$. Instead of $B^{riskless}$, we have the price of the non-vulnerable stock, which, normalized by earnings, is $\left(\frac{P}{E}\right)^{non-vuln.}$.

Second, instead of the recovery rate upon default R, the stock-based equation has the price of the vulnerable stock at default divided by the price of the comparable non-vulnerable stock at default. Normalizing by earnings, the price of the vulnerable stock upon sovereign default is $\left(\frac{P}{E}\right)^{vulnerable} = \frac{1}{\overline{d}-\overline{\mu_x}}$. The normalized price of the comparable non-vulnerable stock is $\left(\frac{P}{E}\right)^{non-vuln.} = \frac{1}{d-\mu_x}$. Therefore, we obtain the ratio $\frac{d-\mu_x}{\overline{d}-\overline{\mu_x}}$ in place of R. Note that we can write:

$$\overline{d} - \overline{\mu_x} = d - \mu_x + \left(\overline{d} - d\right) + \left(\mu_x - \overline{\mu_x}\right) = \left(\frac{E}{P}\right)^{non-vuln.} + K_0$$

After some algebra, we get $\frac{K_0}{\left(\frac{E}{P}\right)^{non-vuln.} + K_0}$ in the stock-based equation in place of $(1 - R)$ in the bond-based equation.

Finally, we need to move from the bond-based risk-neutral default probability Q_t to a stock-based risk-neutral default probability. Andrade (2009) shows that raising Q_t to the power of K_1 (defined in Equation (4)) is the correct adjustment given model assumptions. That is, the stock-based equation has $Q_t^{K_1}$ in place of Q_t.

Collecting what we have thus far:

$$\begin{aligned} &\mathit{Bond\ Equation}: & B_t = &\quad B^{riskless}\left[1-(1-R)\,Q_t\right] \\ &\mathit{Stock\ Equation}: \quad \left(\tfrac{P}{E}\right)_t^{vulnerable} & = &\left(\tfrac{P}{E}\right)^{non-vuln.}\left[1-\frac{K_0}{\left(\frac{E}{P}\right)^{non-vuln.}+K_0}Q_t^{K_1}\right] \end{aligned}$$

Re-arranging the *Stock Equation* gives Equation (5) in the text:

$$VD_t \equiv \frac{\left(\frac{P}{E}\right)^{non-vuln.}-\left(\frac{P}{E}\right)_t^{vulnerable}}{\left(\frac{P}{E}\right)^{non-vuln.}} = \frac{K_0}{\left(\frac{E}{P}\right)^{non-vuln.}+K_0}Q_t^{K_1}.$$

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TABLE 1: SUMMARY STATISTICS

The table has summary statistics. The sample consists of non-financial Eurozone firms from July 2005 to December 2013. The smallest stocks collectively adding up to 1% of total Eurozone market capitalization are excluded. Stock returns are denominated in Euros and from Datastream. Ten-year Euro-denominated sovereign CDS spreads are from Markit. *ΔSpread* is normalized change in sovereign CDS spread as per the following formula:

$$\Delta Spread_t = \frac{Spread_t - Spread_{t-1}}{r_{F,t-1} + Spread_{t-1}}\left(1 - \frac{1}{\left(1 + r_{F,t-1} + Spread_{t-1}\right)^{10}}\right)$$

where r_F is the German 10-year benchmark yield minus the German 10-year CDS spread. Assets (in millions) are winsorized at the 1% level. *Low Tangibility* is a dummy variable flagging firms with asset tangibility in the lowest tercile in a given month. Asset tangibility is defined as Property Plant and Equipment scaled by assets using data from Worldscope. *High EF Dependence* is a dummy variable that flags firms in the highest tercile of the Rajan-Zingales external finance dependence measure from Claessens and Laeven (2003). *Young* is dummy variable flagging firms incorporated after 2001 using incorporation year from OSIRIS. Data are at the monthly frequency.

Country	N (firm-months)	Excess Stock Return		ΔSpread		Assets		Low Tangibility	High EF Dependence	Young
		Mean	Std Dev	Mean	Std Dev	Mean	Median	Mean	Mean	Mean
Austria	3277	0.38%	12.63%	0.03%	1.53%	2935	1000	0.116	0.150	0.033
Belgium	5018	0.60%	0.10%	0.05%	1.55%	3385	848	0.355	0.408	0.065
Finland	5761	0.38%	10.74%	0.01%	0.70%	2729	1026	0.341	0.311	0.137
France	22179	0.40%	10.41%	0.04%	1.36%	9555	1134	0.424	0.318	0.036
Germany	20612	0.73%	11.08%	0.02%	0.73%	8481	711	0.334	0.353	0.050
Greece	4573	-0.57%	13.03%	1.76%	12.29%	1393	562	0.107	0.271	0.000
Ireland	2443	0.60%	13.57%	0.12%	3.24%	2365	943	0.257	0.389	0.127
Italy	10954	0.00%	11.00%	0.15%	2.48%	6656	1033	0.356	0.278	0.121
Netherlands	6341	0.54%	10.47%	0.03%	0.80%	9536	1460	0.414	0.355	0.043
Portugal	2259	0.14%	9.99%	0.38%	4.83%	4605	2096	0.182	0.322	0.083
Spain	7610	0.03%	10.77%	0.16%	2.22%	8551	1395	0.256	0.334	0.042
All Countries	91027	0.36%	11.01%	0.15%	3.26%	7141	991	0.334	0.323	0.054

TABLE 2: DO LOW TANGIBILITY FIRMS HAVE HIGHER SOVEREIGN RISK BETAS?

The table shows results of panel regression of stock-level excess returns onto contemporaneous changes on sovereign spreads and four factors. Data are monthly from July 2005 to December 2013. *Low Tangibility* is a dummy variable flagging firms whose asset tangibility is in the lowest tercile. Standard errors are clustered at the country-time level. T-statistics are reported in parentheses below the coefficients. *,**,*** denote significance at the 10%, 5% and 1% levels respectively.

Dependent Variable:	Excess Stock Returns				
			Industry Fixed Effects	2010-2013	GIIPS only
	(1)	(2)	(3)	(4)	(5)
ΔSpread	-0.135 ***	-0.120 ***	-0.119 ***	-0.122 ***	-0.106 **
	(3.28)	(2.97)	(2.96)	(3.11)	(2.56)
ΔSpread × Low Tangibility		-0.104 ***	-0.103 ***	-0.119 ***	-0.087 ***
		(2.68)	(2.67)	(3.13)	(2.63)
Low Tangibility		0.000	0.001	0.003 ***	-0.001
		(0.36)	(1.15)	(2.74)	(0.33)
MKT	1.021 ***	1.010 ***	1.010 ***	0.988 ***	0.968 ***
	(43.34)	(41.39)	(41.48)	(24.92)	(23.92)
HML	-0.231 ***	-0.221 ***	-0.221 ***	-0.144 ***	-0.09
	(6.50)	(5.96)	(5.95)	(2.79)	(1.24)
SMB	0.384 ***	0.355 ***	0.355 ***	0.355 ***	0.307 ***
	(9.49)	(8.45)	(8.43)	(6.71)	(5.01)
WML	-0.222 ***	-0.216 ***	-0.216 ***	-0.231 ***	-0.303 ***
	(9.06)	(8.96)	(8.93)	(8.48)	(7.36)
MKT × Low Tangibility		0.030	0.028	-0.044	0.049
		(1.18)	(1.19)	(1.13)	(1.31)
HML × Low Tangibility		-0.034	-0.033	-0.018	-0.044
		(0.86)	(0.85)	(0.35)	(0.63)
SMB × Low Tangibility		0.092 **	0.092 **	-0.026	0.037
		(2.59)	(2.56)	(0.56)	(0.58)
WML × Low Tangibility		-0.015	-0.015	-0.002	0.017
		(0.63)	(0.62)	(0.07)	(0.32)
Constant	0.002 **	0.002 *	-0.001	-0.001	-0.0004
	(2.01)	(1.84)	(0.44)	(0.40)	(0.22)
N	91027	91027	91027	51725	27790
R^2	0.220	0.221	0.222	0.189	0.234

TABLE 3: DO LOW-TANGIBILITY FIRMS HAVE HIGHER SOVEREIGN RISK BETAS? (PORTFOLIO RESULTS)

The table shows regressions of stacked country-level long-short portfolio returns on contemporaneous changes in sovereign spreads and four factor returns. Long-short portfolios within each Eurozone country are based on *Low Tangibility* and are equally-weighted in columns (1)-(3) and value-weighted in columns (4)-(6). *Low Tangibility* is a dummy variable flagging firms whose asset tangibility is in the lowest tercile. We report White, Driscoll-Kraay (4 lags), and time-clustered standard errors. T-statistics are in parentheses below the coefficients. *,**,*** denote significance at the 10%, 5% and 1% respectively.

Dependent Variable:	Return (Low Tangibility = 1) - Return (Low Tangibility = 0)					
	Equally-weighted			Value-weighted		
	(1)	(2)	(3)	(4)	(5)	(6)
ΔSpread	-0.049	-0.063	-0.078	-0.055	-0.062	-0.073
Robust	(2.41) **	(3.14) ***	(3.49) ***	(2.38) **	(2.55) **	(2.77) **
Driscoll Kraay	(2.02) *	(3.45) ***	(3.76) ***	(2.11) **	(2.44) **	(2.71) **
Time-clustered	(2.26) **	(2.94) ***	(3.24) ***	(2.12) **	(2.38) **	(2.59) **
MKT		0.028			0.017	
Robust		(0.86)			(0.39)	
Driscoll Kraay		(0.90)			(0.37)	
Time-clustered		(0.94)			(0.34)	
HML		-0.107			-0.080	
Robust		(2.04) **			(1.11)	
Driscoll Kraay		(3.13) ***			(1.00)	
Time-clustered		(2.34) **			(0.93)	
SMB		0.134			0.250	
Robust		(2.48) **			(3.66) ***	
Driscoll Kraay		(2.75) **			(3.98) ***	
Time-clustered		(2.78) **			(3.39) ***	
WML		0.008			-0.093	
Robust		(0.26)			(2.34) **	
Driscoll Kraay		(0.22)			(1.87) *	
Time-clustered		(0.28)			(2.23) **	
Interactions Four Factors × Country Dummies	No	No	Yes	No	No	Yes
N	1092	1092	1092	1092	1092	1092

TABLE 4: DO FIRMS WITH HIGH EXTERNAL FINANCE DEPENDENCE OR YOUNG FIRMS HAVE HIGHER SOVEREIGN RISK BETAS?

The table shows results of panel regression of stock-level excess returns on contemporaneous changes on sovereign spreads and four factors. Data are monthly from July 2005 to December 2013. *High EF Dependence* is a dummy variable flagging firms in the highest tercile of the Rajan-Zingales external finance dependence measure. *Young* is a dummy variable flagging firms incorporated after 2001. Standard errors are clustered at the country-time level. T-statistics are reported in parentheses below the coefficients. *,**,*** denote significance at the 10%, 5% and 1% levels respectively.

Dependent Variable:	Excess Stock Returns			
	High EF Dependence		Young	
		Industry FE		Industry FE
	(1)	(2)	(3)	(4)
ΔSpread	-0.118 ***	-0.116 ***	-0.122 ***	-0.121 ***
	(2.88)	(2.87)	(3.02)	(3.02)
ΔSpread × High EF Dependence	-0.086 ***	-0.087 ***		
	(3.40)	(3.47)		
High EF Dependence	0.001	0.002 **		
	(1.15)	(1.96)		
ΔSpread × Young			-0.159 *	-0.156 *
			(1.74)	(1.72)
Young			-0.002	-0.003
			(1.20)	(1.30)
MKT	1.046 ***	1.046 ***	1.023 ***	1.023 ***
	(41.69)	(41.57)	(41.25)	(41.29)
HML	-0.223 ***	-0.223 ***	-0.241 ***	-0.241 ***
	(5.63)	(5.63)	(6.26)	(6.25)
SMB	0.386 ***	0.384 ***	0.384 ***	0.384 ***
	(9.52)	(9.38)	(9.11)	(9.08)
WML	-0.235 ***	-0.235 ***	-0.217 ***	-0.217 ***
	(8.65)	(8.65)	(8.74)	(8.71)
MKT × High EF Dependence	-0.107 ***	-0.106 ***		
	(4.69)	(4.70)		
HML × High EF Dependence	-0.024	-0.023		
	(0.60)	(0.58)		
SMB × High EF Dependence	-0.002	0.000		
	(0.04)	(0.00)		
WML × High EF Dependence	0.043	0.044		
	(1.59)	(1.63)		
MKT ×Young			-0.001	-0.001
			(0.01)	(0.02)
HML × Young			-0.13	-0.131
			(1.63)	(1.64)
SMB × Young			0.136 *	0.135 *
			(1.85)	(1.84)
WML × Young			-0.073	-0.073
			(1.35)	(1.36)
Constant	0.001	-0.002	0.002 **	-0.002
	(1.50)	(0.77)	(2.27)	(0.72)
N	90769	90769	77871	77871
R^2	0.219	0.219	0.219	0.219

TABLE 5: DO VULNERABLE FIRMS HAVE HIGHER SENSITIVITY TO SOVEREIGN SPREADS FOLLOWING POLICY ANNOUNCEMENTS?

Table 5 has an analysis of two-day returns following three policy announcements pertaining to intervention in sovereign debt markets. Panel A has summary statistics of such announcements. Panel B has panel regressions of stock returns on contemporaneous changes in sovereign spreads and interactions with *Low Tangibility*. *Low Tangibility* is a dummy variable flagging firms whose asset tangibility is in the lowest tercile. Betas with respect to the four factors are estimated using monthly data and in the full sample period of June/2005 to December/2013. In Panel B, t-statistics based on country-clustered standard errors are displayed below coefficients. *,**,*** denote significance at the 10%, 5% and 1% levels respectively.

Panel A: Announcement Returns

Event	Announcement Date	N (firms)	Avg. Stock Return	Avg. ΔSpread
EFSF Creation	May 8, 2010	907	0.0472	-0.0418
Draghi's speech	July 25, 2012	795	0.0381	-0.0174
OMT Creation	September 6, 2012	787	0.0231	-0.0167

Panel B: Regression Results

Dependent Variable:	Stock Returns			
				GIIPS only
	(1)	(2)	(3)	(4)
ΔSpread	-0.132 ***	-0.109 ***	-0.084 **	-0.064 ***
	(3.60)	(3.41)	(2.60)	(2.44)
ΔSpread × Low Tangibility		-0.130 **	-0.107 **	-0.092 **
		(2.20)	(2.55)	(2.48)
Low Tangibility		-0.005 ***	0.000	-0.001
		(3.19)	(0.17)	(0.20)
β_{MKT}			0.029 ***	0.032 ***
			(11.73)	(10.25)
β_{SMB}			-0.014 ***	-0.020 ***
			(4.37)	(5.74)
β_{HML}			0.003	0.010 **
			(0.96)	(2.27)
β_{WML}			-0.015 ***	-0.018 ***
			(4.62)	(2.70)
Industry Fixed Effects	No	No	Yes	Yes
Constant	0.033 ***	0.034 ***	0.008	0.021 **
	(14.00)	(13.74)	(1.29)	(2.39)
N	2489	2489	2489	682
R^2	0.019	0.021	0.143	0.150

TABLE 6: DECOMPOSING SOVEREIGN RISK BETAS

Panel A shows results of panel regressions that decompose sovereign risk betas. As in Equation (1) in the paper, firm-level log price changes are decomposed into changes in log EPS expectations (averaging EPS expectations over the nearest three years) and changes in log P/E ratios based on such EPS expectations. In Panel A, log price changes and its components are separately regressed on changes in sovereign spreads, including interactions with *Low Tangibility*. *Low Tangibility* is a dummy variable flagging firms whose asset tangibility is in the lowest tercile. Panel B shows results of panel regressions of changes in EPS expectations (using average EPS expectations over next three years) scaled by stock prices, using three types of price scalings. Data are quarterly from 2005 to 2013. Standard errors are clustered at the country-time level. T-statistics are reported in parentheses below the coefficients. *,**,*** denote significance at the 10%, 5% and 1% levels respectively.

Panel A: Sovereign Risk Beta Decomposition

Dependent Variable:	Change in log prices		Change in log forward P/E ratios		Change in log EPS expectations	
	(1)	(2)	(3)	(4)	(5)	(6)
ΔSpread	-1.309 ***	-1.154 ***	-1.101 ***	-0.954 ***	-0.208 ***	-0.200 **
	(3.40)	(3.18)	(3.50)	(3.28)	(2.09)	(2.01)
ΔSpread × Low Tangibility		-0.749 ***		-0.710 ***		-0.039
		(5.09)		(5.30)		(0.37)
Low Tangibility		-0.003		-0.005		0.002
		(0.68)		(1.08)		(0.64)
Constant	0.001	0.001	0.005	0.006	-0.004	-0.005
	(0.06)	(0.18)	(0.64)	(0.88)	(0.90)	(1.01)
N	25693	25693	25693	25693	25693	25693
R^2	0.086	0.091	0.036	0.039	0.001	0.002

Panel B: Change in EPS expectations

	Change in EPS Expectations, scaled by prices					
Dependent Variable:	Scaling 1: initial price		Scaling 2: last price		Scaling 3: average price	
	(1)	(2)	(3)	(4)	(5)	(6)
ΔSpread	-0.021 **	-0.02 **	-0.056 *	-0.056 *	-0.021 **	-0.021 **
	(2.35)	(2.19)	(1.96)	(1.79)	(2.40)	(2.20)
ΔSpread × Low Tangibility		-0.003		0.002		0.002
		(0.30)		(0.09)		(0.17)
Low Tangibility		0.000		0.002		0.000
		(0.41)		(0.87)		(0.42)
Constant	0.001	0.001	-0.004 ***	-0.005 **	0.000	0.001
	(1.24)	(1.04)	(2.64)	(2.38)	(0.84)	(0.85)
N	26396	26396	26396	26396	26396	26396
R^2	0.001	0.001	0.001	0.001	0.001	0.001

TABLE 7: STRUCTURAL VALUATION MODEL

The variables Value Discount (VD) and E/P ratio for non-vulnerable firms $(E/P)^{non\text{-}vuln}$ are defined at the country-industry level as described in the text. The Risk Neutral Default Probability (Q) is computed from the sovereign spread (S), recovery ratio (R), and risk-free rate (r) as defined in the text. Data are quarterly from 2005 to 2013 and defined at the country-industry level. 2005 to2013. Panel A reports summary statistics. Panel B has results of the maximum likelihood estimation of parameters K_0 and K_1 in the equation:

$$VD = \frac{K_0}{(\frac{E}{P})^{non-vuln.} + K_0} Q^{K_1} + \varepsilon$$

Standard errors are clustered at the country-industry level. T-statistics are reported in parentheses below the coefficients. *,**,*** denote significance at the 10%, 5% and 1% levels respectively.

Panel A: Summary Statistics

	$(E/P)^{non\text{-}vuln.}$	VD	Q	S	R	r
Mean	0.080	0.024	0.081	0.0066	0.396	0.028
Median	0.077	0.038	0.048	0.0038	0.400	0.030
Std Dev.	0.032	0.356	0.094	0.0080	0.018	0.012
N	725	725	725	725	725	725

Panel B: Structural Valuation Model

	Baseline (at least 3 firms)	No filter (one firm allowed)	At least 2 firms	At least 4 firms	GIIPS only (at least 3 firms)
K_0 (Cost of Default)	0.0158 ***	0.0134 ***	0.0145 ***	0.0161 ***	0.0150 **
	(3.48)	(3.89)	(3.71)	(3.53)	(2.17)
K_1	0.269 **	0.335 **	0.351 **	0.250 **	0.312
	(2.61)	(2.57)	(2.32)	(2.43)	(1.25)
Corr(VD^{fitted}, VD)	0.48	0.29	0.38	0.53	0.54
N	725	1472	1021	576	234

APPENDIX TABLE A1: DOES LOW TANGIBILITY CAPTURE CREDIT VULNERABILITY?

Panel A shows summary statistics of change in bank loans/assets, return spread and *Low Tangibility* from 2005-2013. Data are annual and from OSIRIS. Sovereign spreads are measured at the end of each calendar year. *Low Tangibility* is a dummy variable flagging firms whose asset tangibility is in the lowest tercile. Panel B shows panel regression results of changes in firm-level bank loans/assets on changes in sovereign spreads, including interactions with *Low Tangibility*.. Columns (4) to (6) include 2-digit SIC industry fixed effects. Standard errors are clustered at the firm level. T-statistics are shown below coefficient estimates. *,**,*** denote significance at the 10%, 5% and 1% levels.

Panel A: Summary Statistics

	ΔBankLoans/Assets	ΔSpread	Low Tangibility
Mean	0.012	0.008	0.248
Median	-0.001	0.001	0.000
Std Dev.	0.105	0.222	0.432
N	12230	12230	12230

Panel B: Panel Regressions

Dependent Variable:	Δ Bank Loans/Assets		
	(1)	(2)	(3)
ΔSpread	-0.021 ***	-0.016 ***	-0.016 ***
	(4.73)	(3.75)	(3.70)
ΔSpread × Low Tangibility		-0.045 **	-0.047 **
		(2.15)	(2.20)
Low Tangibility		-0.003	-0.003
		(1.45)	(1.09)
Constant	0.012 ***	0.011 ***	0.013 ***
	(13.58)	(11.61)	(12.07)
Industy Fixed Effects (SIC -2digit)	N	N	Y
N	12230	12230	12230
R^2	0.002	0.003	0.009

APPENDIX TABLE A2: INDUSTRY-COUNTRY DISTRIBUTION OF LOW TANGIBILITY AND OTHER FIRMS

The table shows the industry-country distribution of firms used in the structural estimation model. The top number represents the total number of time-series observations (quarters), the middle number is the average number of *Low Tangibility*=1 firms, and the bottom number is the average number of *Low Tangibility*=0 firms in each industry-country group. Data are quarterly and the sample period is from 2005 to 2013.

Country \ Industry	Oil & Gas	Basic Materials	Industrials	Consumer Goods	Health Care	Consumer Services	Telecom	Utilities	Technology
Austria			35 1.5 11.6			16 1.0 1.3			
Belgium			35 3.1 8.3	29 1.5 5.1	23 3.4 1.4			11 1.0 1.2	35 2.9 1.0
Finland		9 1.0 7.3	35 9.1 12.2	35 2.0 6.5	27 1.0 1.1	31 1.4 5.3			14 6.5 1.2
France	23 1.0 6.3	21 1.2 8.2	35 22.5 23.9	35 13.9 24.2	35 3.2 12.6	35 17.7 19.1	12 1.3 1.3		35 24.1 3.4
Germany	35 3.1 6.8	5 1.0 10.6	35 17.8 34.4	35 4.8 20.2	35 5.0 7.5	35 8.7 13.9	35 1.8 1.1		35 15.1 6.2
Greece			18 2.2 7.7	7 2.1 4.3		30 2.3 6.3			
Ireland			28 1.0 4.3	35 2.3 4.4		30 1.3 4.8			
Italy		3 1.0 3.0	35 11.3 17.6	35 5.6 13.6		35 7.9 4.5			35 4.4 2.0
Netherlands		4 1.0 2.0	33 11.9 12.7	21 1.5 7.0	20 1.6 1.2	33 4.8 5.7			29 5.6 2.1
Portugal			20 2.0 6.5			28 2.4 4.9			
Spain	35 1.9 3	7 1.0 9.7	35 5.5 11.3	35 2.1 8.7	32 1.6 3.8	35 3.8 6.2			

APPENDIX TABLE A3: LINEAR REGRESSION USING STRUCTURAL MODEL INPUTS

The variables Value Discount (VD) and E/P ratio for non-vulnerable firms $(E/P)^{non\text{-}vuln}$ are defined at the country-industry level as described in the text. The Risk Neutral Default Probability (Q) is computed from the sovereign spread (S), recovery ratio (R), and risk-free rate (r) as defined in the text. Data are quarterly from 2005 to 2013 and defined at the country-industry level. The table reports linear regressions of VD on Q. Standard errors are clustered at the country-industry level. T-statistics are reported in parentheses below the coefficients. *,**,*** denote significance at the 10%, 5% and 1% levels respectively.

Dependent Variable: VD		GIIPS only
$(E/P)^{non\text{-}vuln.}$	-6.542 ***	-6.742 ***
	(7.34)	(4.88)
Q	0.719 **	0.616 *
	(2.09)	(1.78)
Constant	0.492 ***	0.461 ***
	(6.94)	(4.73)
R^2	0.36	0.44
N	725	725

FIGURE 1: EUROPEAN CDS SPREADS

The Figure shows 10-year Eurozone sovereign CDS spreads on a monthly basis from 2004-2013. Data are Euro-denominated and based on contracts with the Full Restructuring clause. Data are from Markit. Greek sovereign CDS data ends in February 2012 and is capped at 0.13.

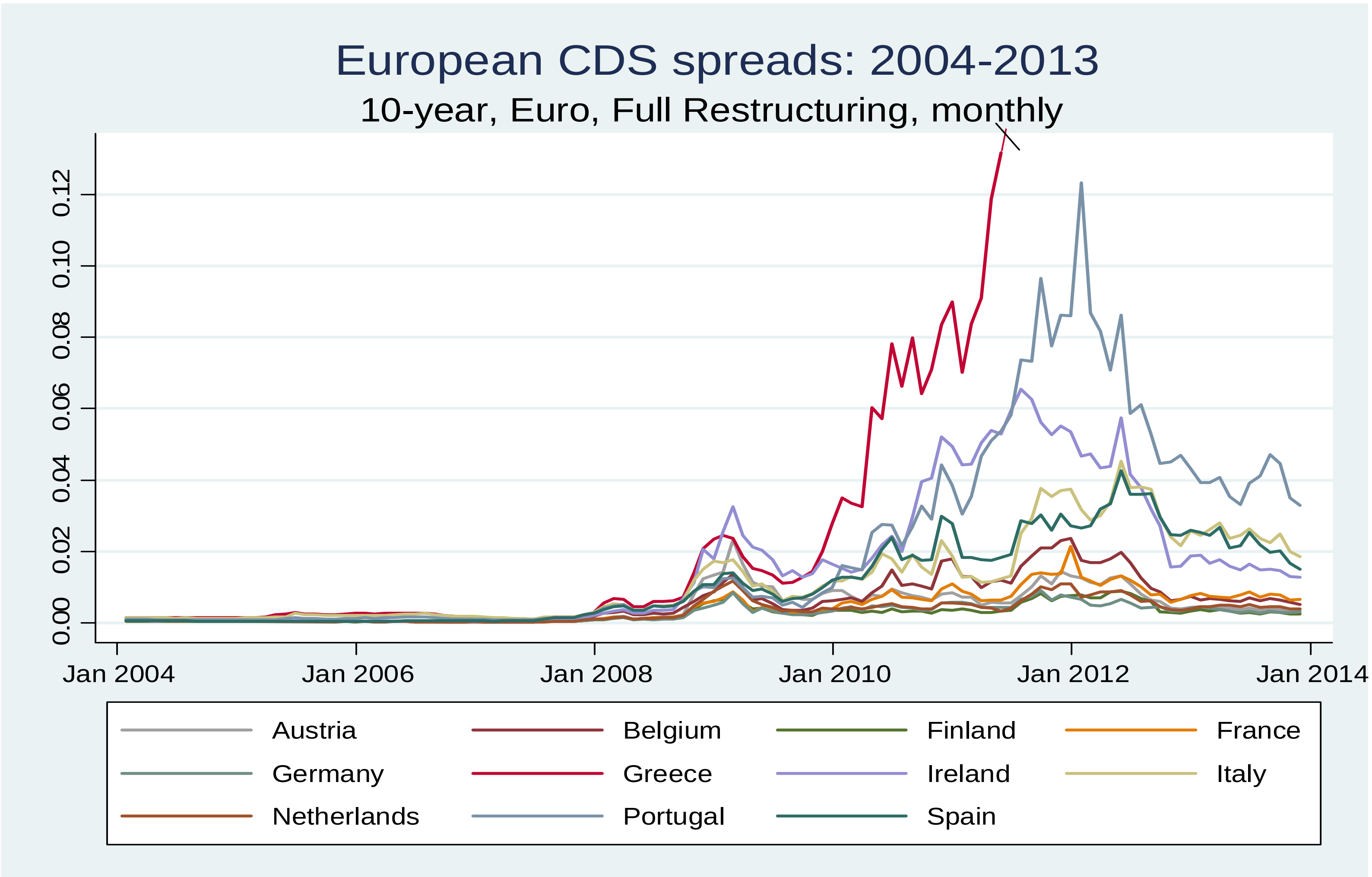

FIGURE A.1: EQUITY MARKET SEGMENTATION

The Figure shows Bekaert et al. (2011) measure of equity market segmentation. The measure is based on pairwise absolute differences of industry-level earnings yields. Earnings yields are computed using average earnings-per-share forecasts over the nearest three years. We show two measures: one from Europe, and the other one from the Rest of the Developed World (ROW). Data are monthly from January 1987 to December 2013.